Belief Space Partitioning for Symbolic Motion Planning

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AUV marine autonomy

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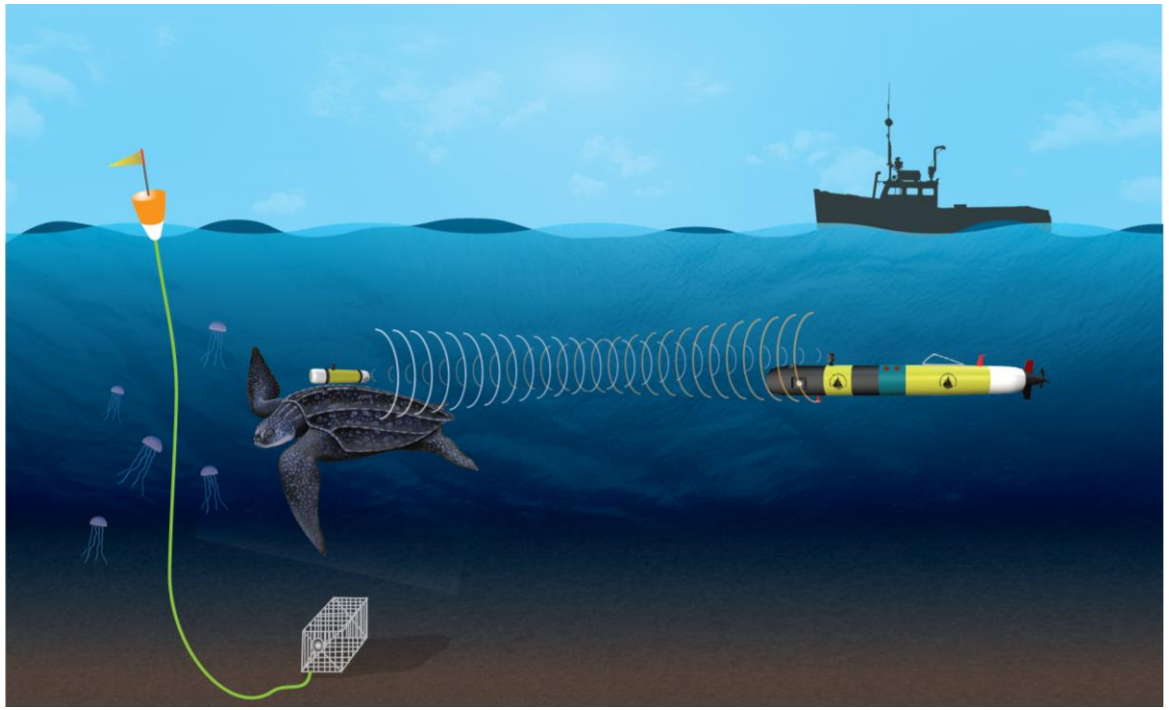
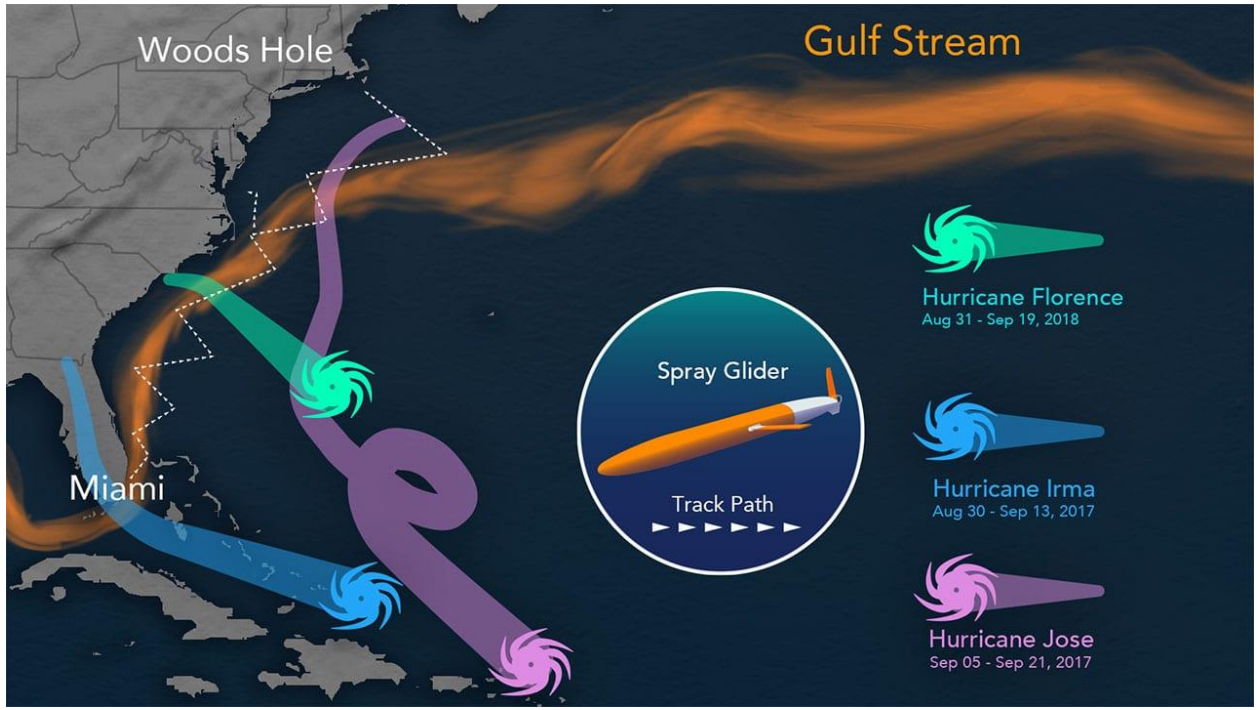


Localize & track hotspots (courtesy Woods Hole)



Ocean sampling (courtesy Woods Hole)

Challenge in marine autonomy

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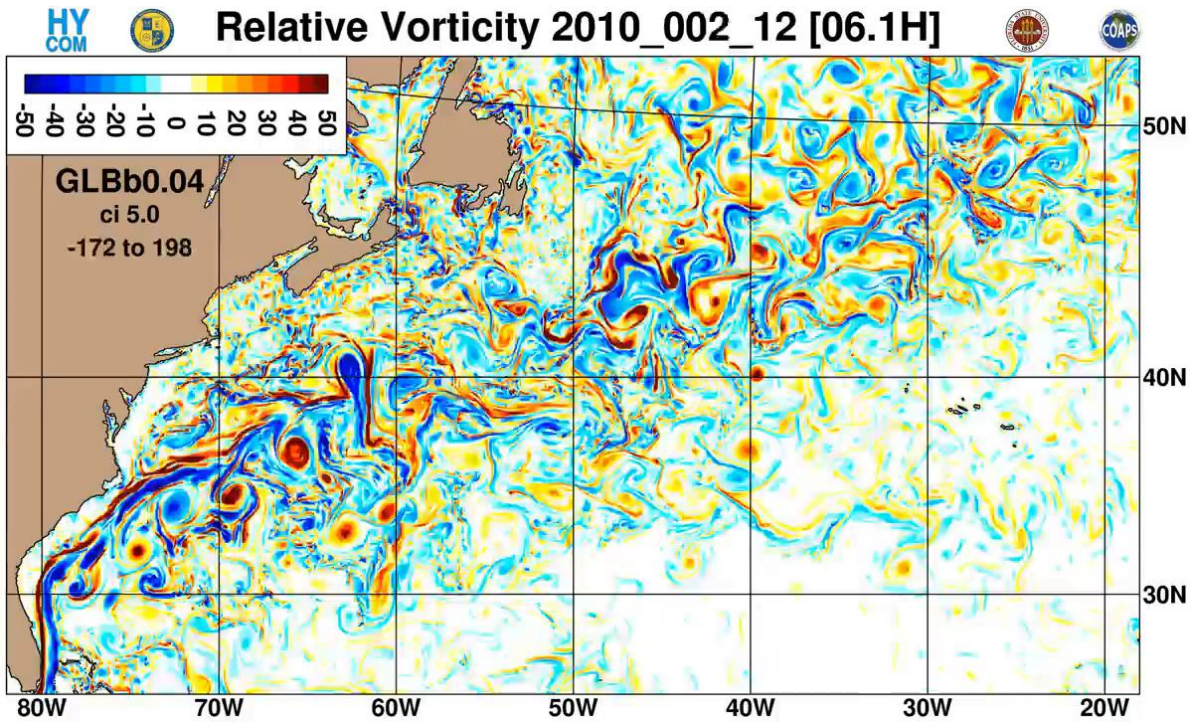
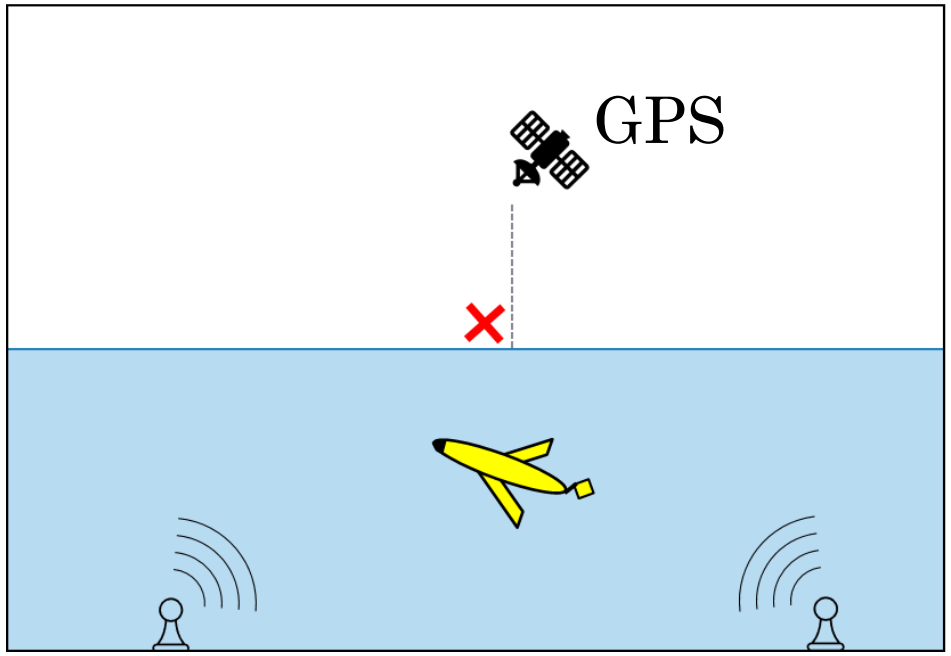


Flow forecasts can contain high uncertainty



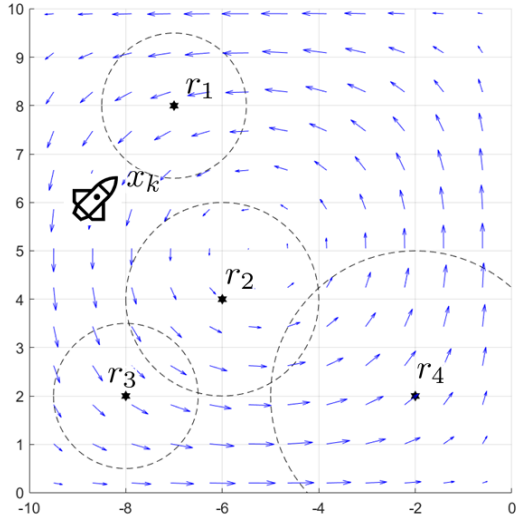
Acoustic localization contains high uncertainty due to multi-path effects and doppler spread

Problem formulation

Vehicle dynamics:

$$x_{k+1} = x_k + F(x_k, t_k) + \begin{bmatrix} V(\cos(u_k^{(1)}) - \frac{\sqrt{2}}{2}|u_k^{(2)}|) \\ V(\sin(u_k^{(1)}) - \frac{\sqrt{2}}{2}|u_k^{(2)}|) \\ u_k^{(2)} \end{bmatrix} + w_k$$

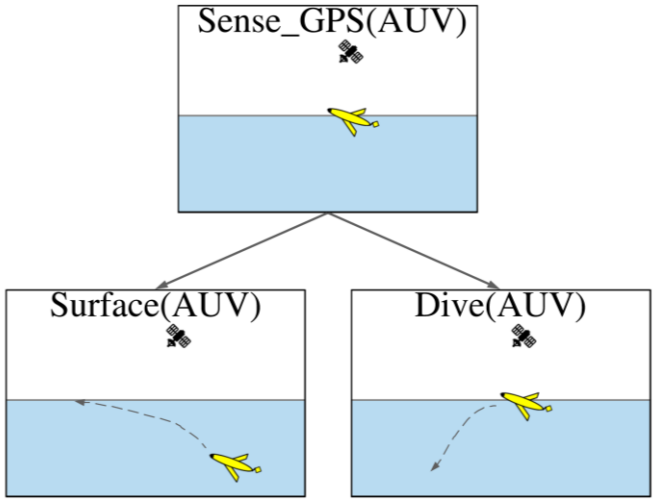
Predicted flow velocity Vehicle forward speed Noise



Control input:

$$u_k \in \mathcal{U}_{xy} \cup \mathcal{U}_z = \{[\tfrac{j\pi}{4} \quad 0]^\intercal\}_{j=0}^{7} \cup \{[\tfrac{\pi}{4} \quad 1]^\intercal, [\tfrac{\pi}{4} \quad -1]^\intercal\}$$

Move Active sensing: diving & surfacing



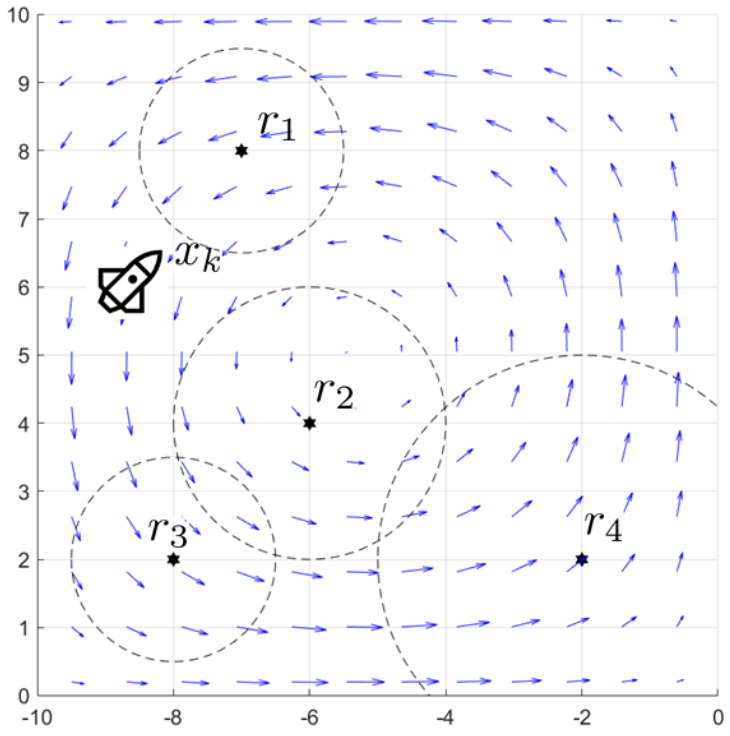
Problem formulation

Localization signal for vehicle: $y_k \in \mathbb{R}^M, y_k^{(j)} \in \{0,1\}$

Observation model:

$$\Pr(y_k^{(j)} = 1) = p\mathbb{1}\{\|x_k - r_j\| \leq R_j\} + (1-p)\mathbb{1}\{\|x_k - r_j\| > R_j\}$$

$$\Pr(y_k^{(j)} = 0) = (1-p)\mathbb{1}\{\|x_k - r_j\| \leq R_j\} + p\mathbb{1}\{\|x_k - r_j\| > R_j\}$$



Problem formulation

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Belief state: $b_k = \Pr(x_k | u_{k-1}, u_{k-2}, \ldots, u_1, y_k, y_{k-1}, \ldots, y_1)$

Push forward operator (Bayes' law)

Belief dynamics: $b_{k+1} = \mathcal{L}(y_{k+1}, u_k) b_k$

Problem 1: find an approximation model of the belief dynamics

Problem 2: solve the planning problem

$$\min_{[u_0, \ldots, u_N]} \sum_{k=1}^{N} l(b_k, u_k)$$

$$\text{s.t. approx. belief dynamics,}$$

$$b_0 = \rho_0, g(b_N) \leq 0$$

Difficulty in modeling the belief dynamics

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Difficulty:

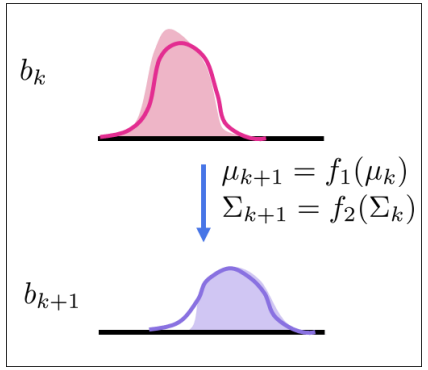
-Nonlinear dynamics, non-Gaussian noise

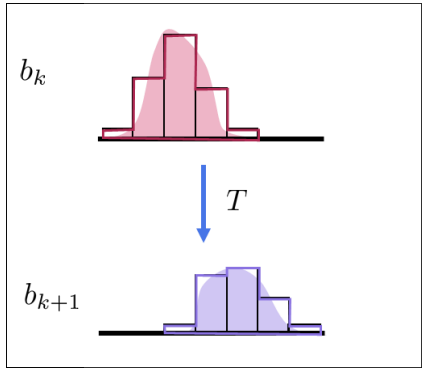
-Curse of dimensionality

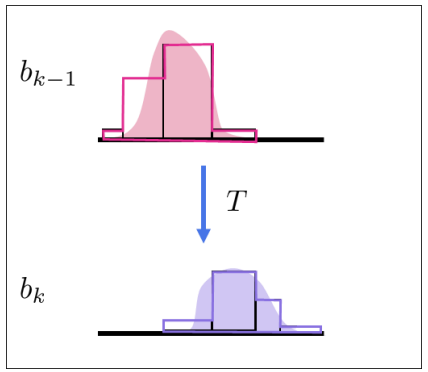
Our contribution:

A novel belief partition method that allocates a fixed number of partitions over regions that the AUV most likely will visit

The partition method enables computationally efficient solutions to the continuous-state POMDP problem







Proposed belief abstraction method

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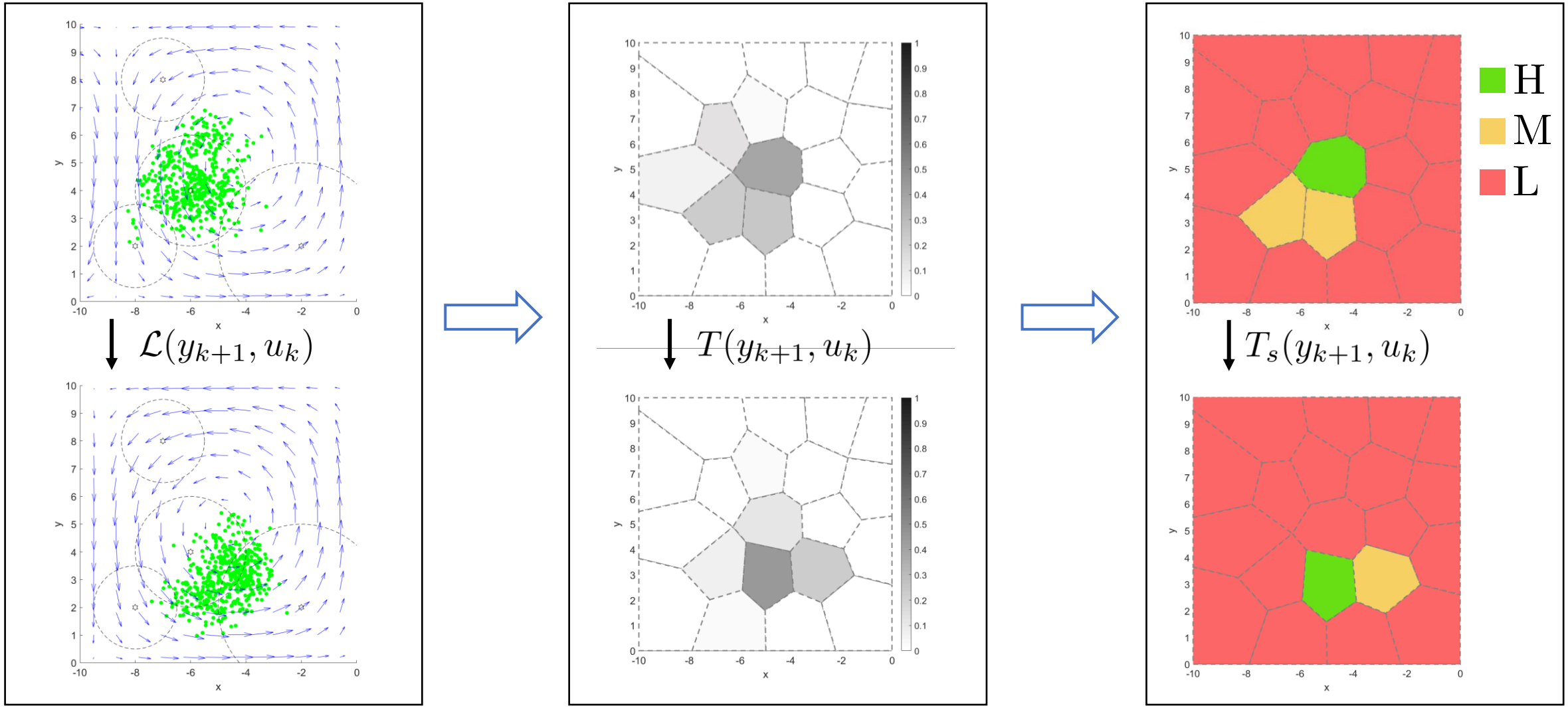
Unparameterized dyn. | Belief dyn. in parameter space | Belief dyn. in symbolic space



Focused finite partition

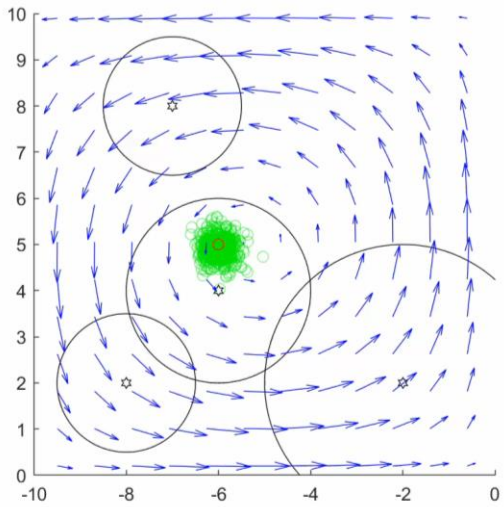
Intuition: in many robotic applications, vehicle has zero probability to be in most of the state space.

We allocate a fixed number of partitions over regions that the system state most likely will visit

Particle position

Simulated particle: $z_k^{(m)} = [(r_k^{(m)})^\intercal, w_k^{(m)}]^\intercal$

Weight



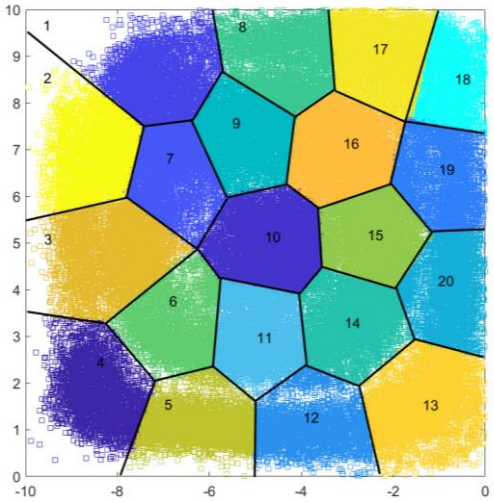
Simulated particle

Centroid of partitioned cells

$$\min_{\mathcal{R}} \sum_{k\in[1,n]} \sum_{i\in[1,K]} \sum_{r_k^m \in \mathcal{R}_i} \text{dist}(z_k^{(m)}, \mu^{(i)}) \qquad (1)$$

$$\text{s.t.} \quad \sum_{i\in[1,K]} I\mu^{(i)} = 1, I\mu^{(i)} > 0, \forall i \in [1,K]$$

(1) can be solved using the K-means method

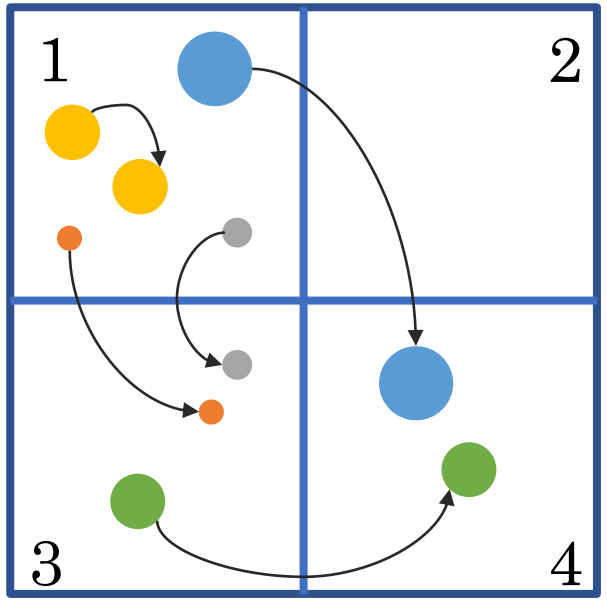


Identify the parameter space belief dynamics

Simulate particle trajectory given a fixed control input u,

Numerically estimate the partitioned belief dynamics by:

$$T^{ij}(u) = \frac{\text{mass transported from } j \text{ to } i}{\text{total mass in } j}$$



Belief dynamics in parameter space:

$$\Theta_k = T(u_{k-1})\Theta_{k-1},$$

$$\Theta_0 = \arg\min_{\Theta} \operatorname{dist}(b_0 - \Theta^T \Phi)$$

Identify the symbolic space belief dynamics

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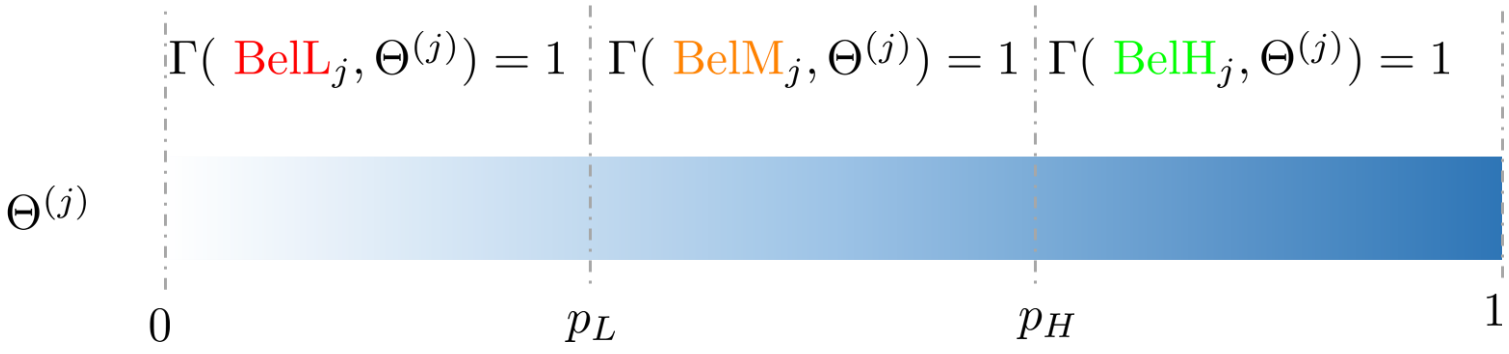
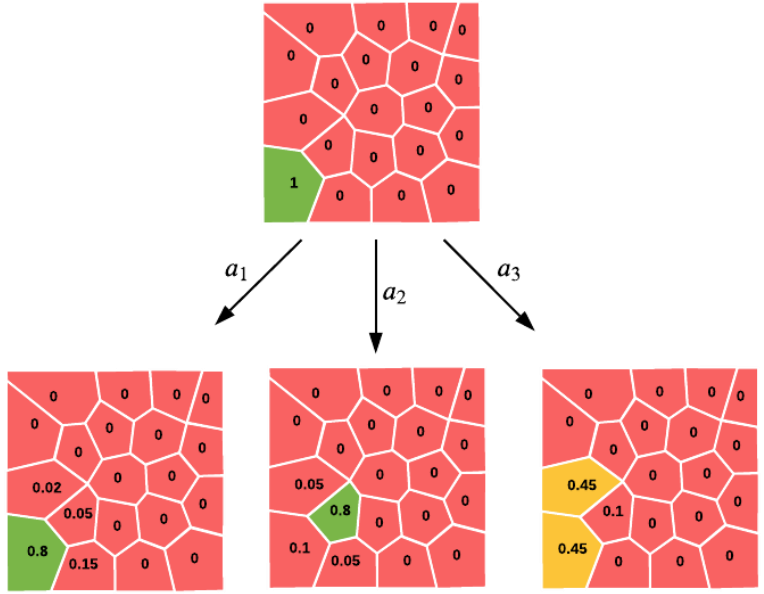
Predicates: $\mathbb{S} = \{\text{at_surf}, \text{at_depth}\} \cup \{\text{BelH}_j, \text{BelM}_j, \text{BelL}_j\}_{j=1}^{K}$

Active predicates: $s_k = \{s \in \mathbb{S} : \Gamma(s, \Theta_k) = 1\}$

Γ defines a mapping from belief values to symbols:

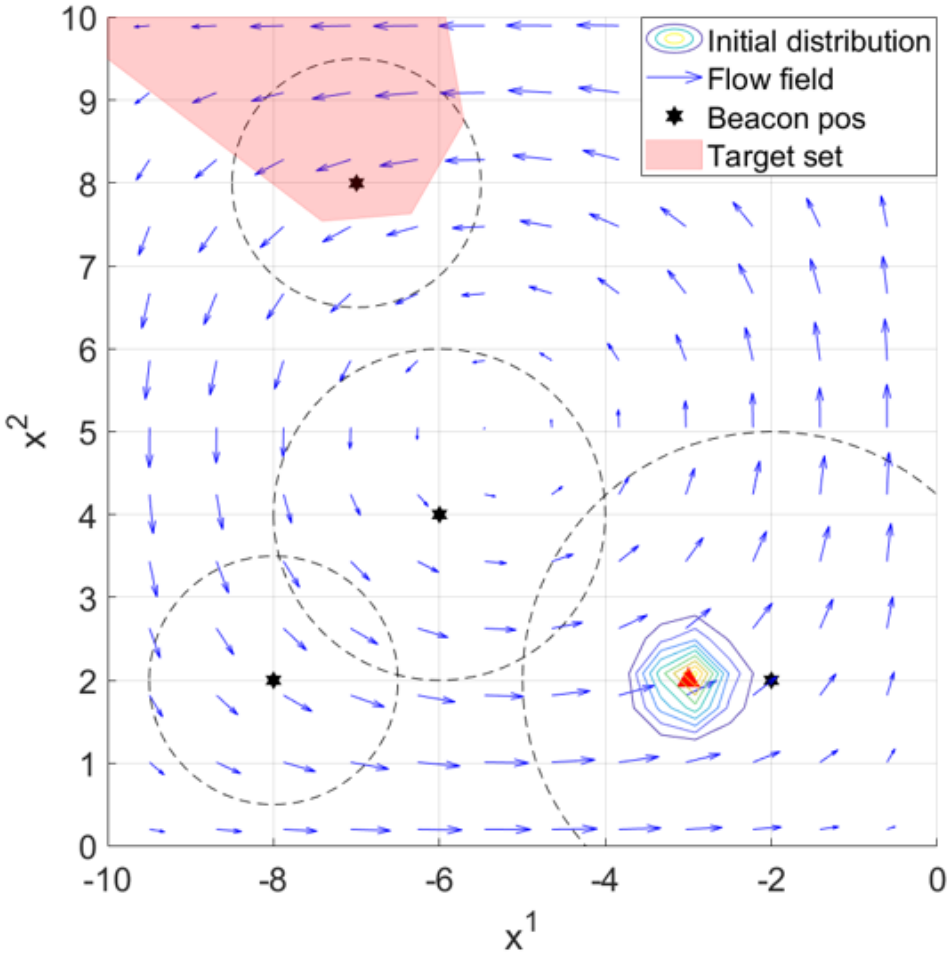




Simulation

We derive the approximated belief dynamics using both the proposed method and the generalized cell mapping method

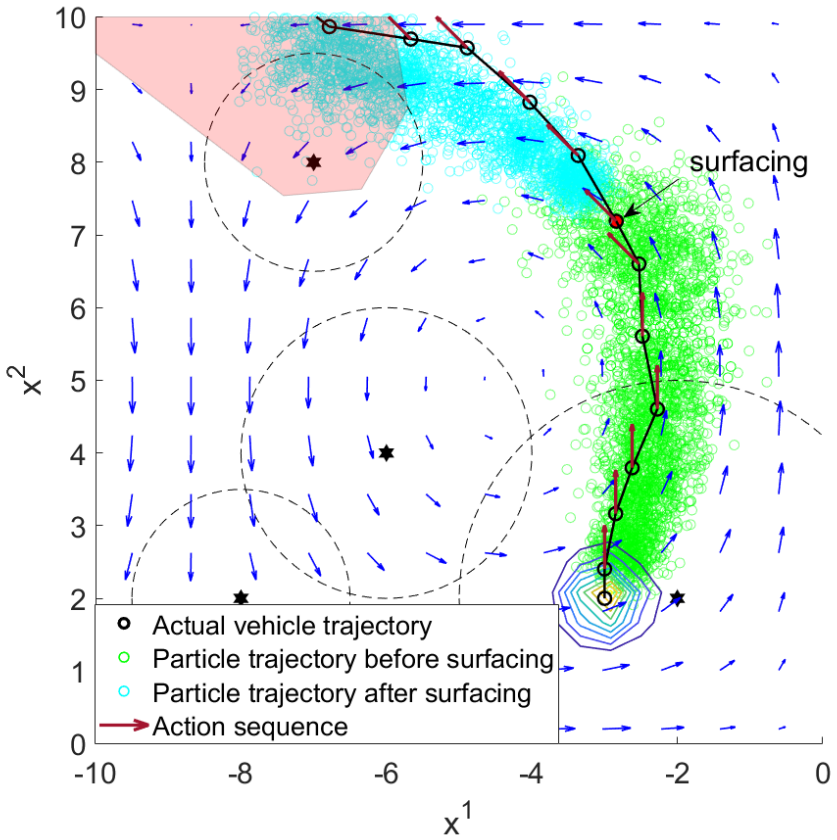
The symbolic planning problem is solved by the A* algorithm



Sun, Jian-Qiao, et al. *Cell Mapping Methods*. Springer, New York, 2018.

Simulation

Generalized cell mapping			
Grid size	# of Pre.	Mod. time	Plan time
$10 \times 10 \times 2$	600	14.2845 s	**5.9873** s
$100 \times 100 \times 2$	$6e4$	15.5396	**54.3957** s
$150 \times 150 \times 2$	$1.35e5$	16.4289 s	**154.368** s
Proposed solution			
Grid size	# of Pre.	Mod. time	Plan time
$10 \times 10 \times 2$	120	18.3886	**3.9702** s
$100 \times 100 \times 2$	120	19.845 s	**4.7825** s
$150 \times 150 \times 2$	120	20.5480 s	**4.9796** s



Belief state partition reduces the computational cost of symbolic planning

Conclusion

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- We propose a novel belief partition method that allocate a fixed number of partitions over regions that the system state most likely will visit

- The partition method enables computationally efficient solutions to the continuous-state POMDP problem